

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF COMPUTING
EXAMINATION

Subject : COMP4431 Artificial Intelligence

Session : 2024 / 2025 Semester I

Date : 17 December 2024

Time : 19:00 - 21:00

Time Allowed: 2 Hours

Subject Examiner(s): Dr LIU Yan

This question paper has a total of 5 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

Answer all the questions in the answer book.

Close book.

A calculator is allowed.

Do not turn over the page until you are told to do so!

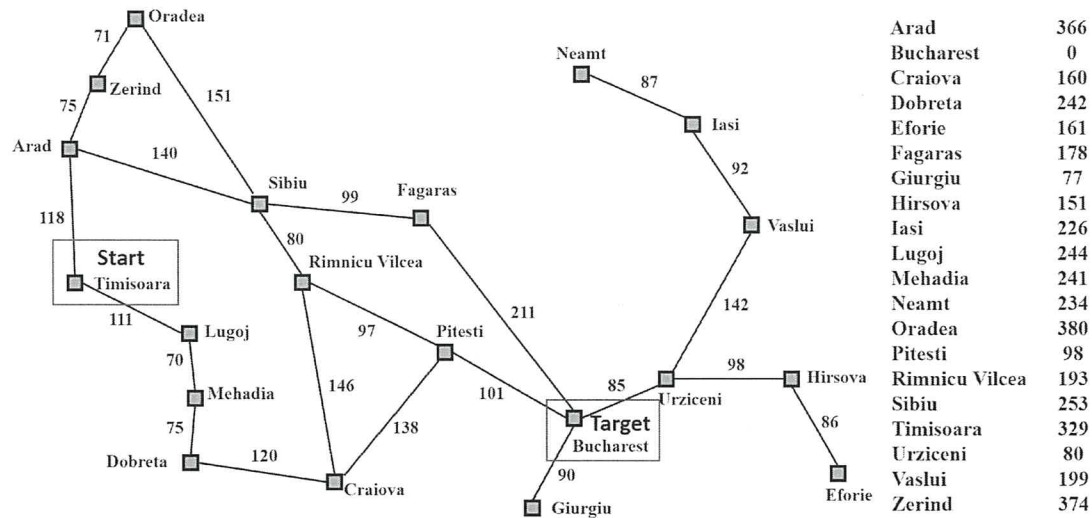
[3]

Section B**[80 Points]**

Question 3. Answer the questions in a short paragraph.

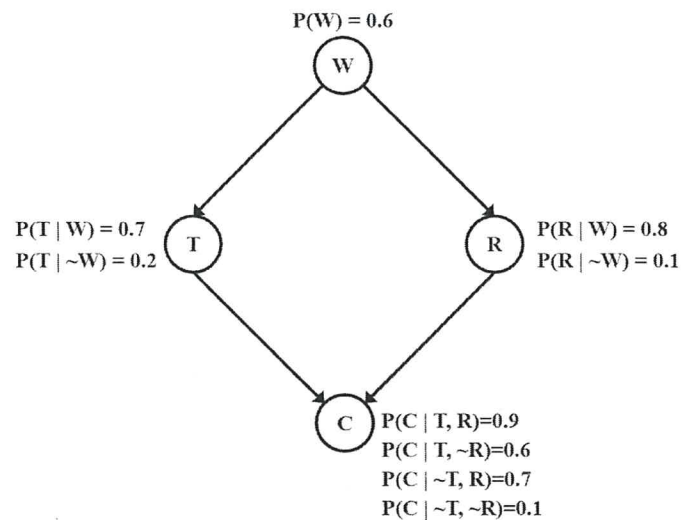
- 3.1. Could you please explain the differences between discriminative AI and generative AI using examples? [6]
- 3.2. Could you please explain the differences between classification and the clustering using examples? [6]
- 3.3. Could you please explain the differences between most constrained variable and least constraining value in constraint satisfaction problem using examples? [6]
- 3.4. Could you please explain the differences between forward chaining and backward chaining using examples? [6]

Question 4. If we want to drive from the city of Timisoara to Bucharest according to the figure below (the number of the distance between two cities has shown in the graph). Could you please find the solution using A* search algorithm (according to the heuristic information of the straight-line distance between any city to the Bucharest). Write down each step of problem solving. [12]



Question 5. Bayesian networks

The Bayesian Network models how weather conditions affect traffic and road conditions, which influences the overall commute time. For example, good weather might lead to dry roads and potentially lighter traffic, while bad weather could result in wet roads and heavier traffic, both of which could contribute to a longer commute time. As shown in the figure below, could you please calculate $P(C)$ and $P(W|C)$? Write down each step of the calculations. [12]



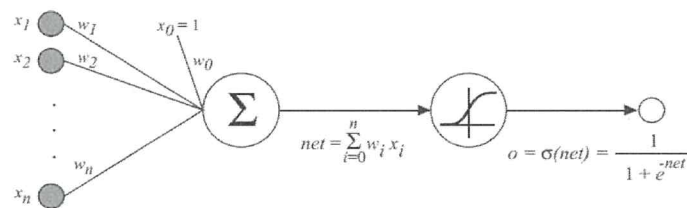
Question 6. Decision trees.

Please construct the decision tree to classify the data points with three attributes into two categories according to the information gain. Please write down all the calculation of entropy and information gain. [12]

Student No.	Study Years	Study Hours	Attendance	Pass (Yes/No)
1	2	Low	I	N
2	2	Low	I	N
3	3	Medium	I	N
4	3	Low	R	N
5	2	High	I	Y
6	2	High	R	Y
7	3	High	R	Y
8	3	Medium	R	Y

Question 7. Neural Networks.

Consider a simple multilayer perceptron with two input nodes, one hidden nodes, and one output node. The activation function is sigmoid as shown below:



while

$$\begin{aligned} \square \quad o_j &= \frac{1}{1 + e^{-net}} = \frac{1}{1 + e^{-\sum_{i=0}^n w_{ij} x_i}} \\ \square \quad \widetilde{y}_p &= \frac{1}{1 + e^{-net}} = \frac{1}{1 + e^{-\sum_{j=0}^M w_{jp} o_j}} \end{aligned}$$

The initial weights from input layer to hidden layer and from hidden layer to output layer are:

$$W(0)^{i-h} = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix} \quad W(0)^{h-o} = \begin{bmatrix} 0.5 \\ 0.6 \end{bmatrix}$$

Bias between input layer and hidden layer, and the bias between hidden layer and output layer are equal to zero.

Given the input $x = [1, 0]$ and the target output $y = 1$. The learning rate η is set to 0.5. Perform one iteration of the backpropagation algorithm and calculate the updated $W(I)^{i-h}$ and $W(I)^{h-o}$ according to the given equations:

- For each output neuro p

$$w_{jp}(k+1) = w_{jp}(k) + \eta \delta_p o_j$$

$$Def. \delta_p = (y_p - \tilde{y}_p) \tilde{y}_p * (1 - \tilde{y}_p)$$
- For each hidden neuro j

$$w_{ij}(k+1) = w_{ij}(k) + \eta \delta_j x_i$$

$$Def. \delta_j = \left(\sum_p w_{jp} \delta_p \right) \tilde{o}_j * (1 - \tilde{o}_j)$$

The number is accurate to 4 decimal places (keep 4 digits after the decimal point) in all calculations. For example, for the number 0.00123456, 0.0012 is used in the calculations and for the number of 1.23456, 1.2345 is used in the calculation. [20]

** END **